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Notes

Simplification of Boolean Expression

(a) $(CB + CC)(A + B + C)$

$= (CB + C)(A + B + C)$ [$\because CC = C$]

$= C(A + B + C)$ [By absorption law]

$= CA + CB + C.C$

$= CA + CB + C$

$= CA + C$

$= C$ //

SOP (Sum of Product)

e.g. $(A \cdot B) + (B \cdot C)$

POS (Product of Sum)

$(A + B) \cdot (B + C)$

(b.) $XY + X'Z + YZ$

$= XY + X'Z + YZ(X + X')$

$= XY + X'Z + XYZ + X'YZ$

$= XY + XYZ + X'Z + X'YZ$

$= XY(1 + Z) + X'Z(1 + Y)$

$= XY + X'Z$ //

(c) $A + B(A + B) + A(A' + B)$

$= A + BA + BB + AA' + AB$

$= A + BA + B + 0 + AB$

$= A + B(A + 1) + AB$

$= A + B + AB$

$= A + B(1 + A) = A + B$ //

(d) $(A+B)(A+C)$

$= AA + AC + BA + BC$

$= A + AC + AB + BC$

$= A(1+C+B) + BC$

$= A.1 + BC$

$= A + BC$ //

(e) $(A+B)(A+\bar{B})(\bar{A}+C)$

$= (AA + A\bar{B} + BA + B\bar{B})(\bar{A}+C)$

$= (A + A\bar{B} + AB + 0)(\bar{A}+C)$

$= A(1+\bar{B}+B)(\bar{A}+C)$

$= A.1(\bar{A}+C)$

$= A\bar{A} + AC$

$= 0 + AC$

$= AC$ //

(f) $\bar{Y}\bar{Z} + \bar{W}\bar{X}\bar{Z} + \bar{W}\bar{X}\bar{Y}\bar{Z} + Y\bar{Z}$

$= \bar{Y}\bar{Z}(1+\bar{W}\bar{X}) + \bar{W}\bar{X}\bar{Z} + Y\bar{Z}$

$= \bar{Y}\bar{Z} + \bar{W}\bar{X}\bar{Z} + Y\bar{Z}$

$= \bar{Z}(\bar{Y}+Y) + \bar{W}\bar{X}\bar{Z}$

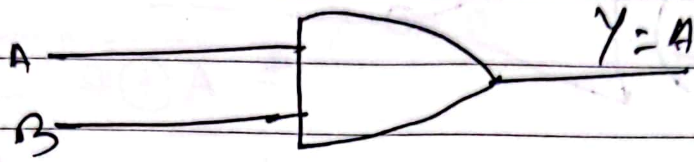
$= \bar{Z}.1 + \bar{W}\bar{X}\bar{Z}$

$= \bar{Z}(1+\bar{W}\bar{X})$

$= \bar{Z}$ //

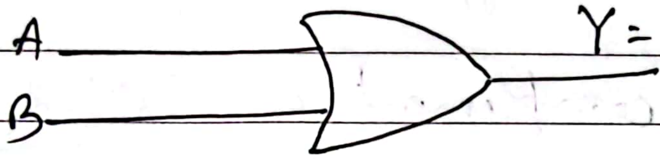
Logic Gates.

1. AND



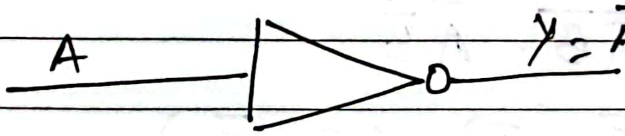
$Y = A \cdot B$

2. OR



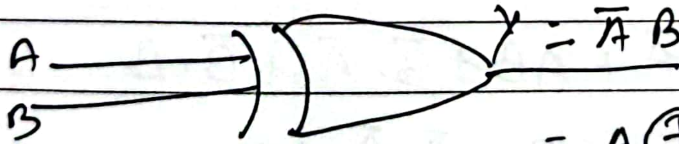
$Y = A + B$

3. NOT (Inverter).



$Y = \overline{A}$

4. EXOR / XOR



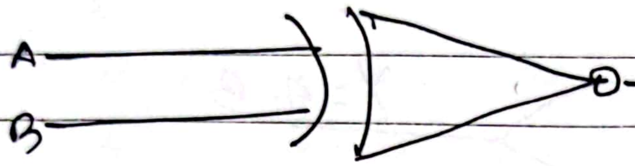
$Y = \overline{A}B + A\overline{B}$

$= A \oplus B$

A	B	$A \oplus B$
1	1	0
1	0	1
0	1	1
0	0	0

If inputs are same, o/p = 0.
If inputs are different o/p = 1.

5. ExNoR / XNoR



$$Y = \bar{A}\,\bar{B} + AB$$
$$= \overline{A \oplus B}$$

If inputs are same, output is 1 and if inputs are different, output is 0.

ExOR is complement of ExNoR Gate.

$$\therefore Y = \overline{\bar{A}B + A\bar{B}}$$
$$= \overline{\bar{A}B} \cdot \overline{A\bar{B}}$$
$$= (\bar{\bar{A}} + \bar{B}) \cdot (\bar{A} + \bar{\bar{B}})$$
$$= (A + \bar{B}) \cdot (\bar{A} + B)$$
$$= A\bar{A} + AB + \bar{B} \cdot \bar{A} + \bar{B} \cdot B$$
$$= 0 + AB + \bar{A}\,\bar{B} + 0$$
$$= AB + \bar{A} \cdot \bar{B}$$ Proved

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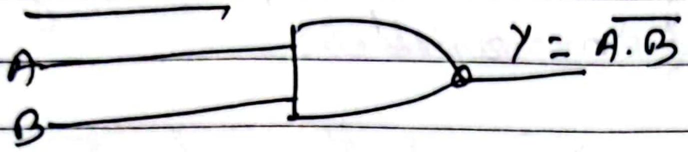
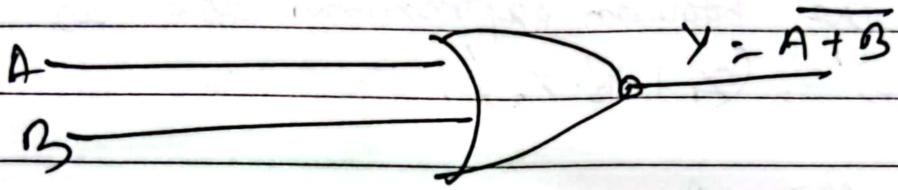
Universal Gates

NAND



NOR



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Boolean Functions:

A function $f: A^n \to A$ is called Boolean Function if it can be specified by a Boolean Expression of 'n' variables.

let $A = \{x_1, x_2, \ldots, x_n\}$ and three operations $+, \cdot, '$.

Then the boolean expressions are as $(x_1 \cdot x_2)'$, $(x_1 + x_2' \cdot x_3)'$, ...

Ex: (i) $f: A^2 \to A$

e.g. $f(x_1, x_2) = x_1 \vee x_2$ or $x_1 \wedge x_2$

(ii) $f: A^3 \to A$

e.g. $f(x_1, x_2, x_3) = (x_1 \vee x_2) \vee x_3 = x_1 \vee (x_2 \vee x_3)$

Types of Boolean Functions

(i) POS form (Product of Sum form)

$(\bar{x}_1 + x_2 + x_3) \cdot (\bar{x}_1 + \bar{x}_2 + x_3) \cdot (x_1 + x_2 + x_3)$

OR

$\underbrace{(\bar{x}_1 \vee x_2 \vee x_3)}_{\text{maxterms}} \wedge \underbrace{(\bar{x}_1 \vee \bar{x}_2 \vee x_3)}_{\text{maxterms}} \wedge \underbrace{(x_1 \vee x_2 \vee x_3)}_{\text{maxterms}}$

CNF: A Boolean expression is said to be conjunctive normal form (CNF) if it is meet of maxterms.

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(ii) SOP Form (Sum of Product form)

$$(\bar{x}_1 \cdot x_2 \cdot x_3) + (\bar{x}_1 \cdot \bar{x}_2 \cdot x_3) + (x_1 \cdot x_2 \cdot x_3)$$

OR

$$(\bar{x}_1 \wedge x_2 \wedge x_3) \vee (\bar{x}_1 \wedge \bar{x}_2 \wedge x_3) \vee (x_1 \wedge x_2 \wedge x_3)$$

Disjunctive Normal form : A Boolean expression is said to be DNF if it is join of minterms.

Que: let $f = (\bar{x} \wedge z) \vee y$. Write f in DNF (join of minterms) or minterm normal forms.

x	y	z	$\bar{x}$	$\bar{x} \wedge z$	f	
1	1	1	0	0	1	(m_1)
1	1	0	0	0	1	(m_2)
1	0	1	0	0	0	(m_3)
1	0	0	0	0	0	(m_4)
0	1	1	1	1	1	(m_5)
0	1	0	1	0	1	(m_6)
0	0	1	1	1	1	(m_7)
0	0	0	1	0	0	(m_8)

अब join of minterms को last column मा 1 भएको value लाई consider गर्ने। यहाँ m_1, m_2, m_5, m_6, m_7 को value 1 छ। त्यसैले m_1 लाई minterms बनाउँदा $m_1 = x \wedge y \wedge z$ लेख्नुपर्छ। त्यस्तै $m_2 = x \wedge y \wedge \bar{z}$, $m_5 = \bar{x} \wedge y \wedge z$, $m_6 = \bar{x} \wedge y \wedge \bar{z}$, $m_7 = \bar{x} \wedge \bar{y} \wedge z$

$$\therefore \text{DNF} = (x \wedge y \wedge z) \vee (x \wedge y \wedge \bar{z}) \vee (\bar{x} \wedge y \wedge z) \vee (\bar{x} \wedge y \wedge \bar{z}) \vee (\bar{x} \wedge \bar{y} \wedge z)$$

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त्यसले CNF पत्ता लगाउन last Column मा 0 भएको value लाई consider गर्नै। यहाँ m_3, m_4 र m_8 को value 0 छ। त्यसैले यिनीहरूलाई maxterm बनाउन निम्नानुसार लेख्नुपर्छ।

$m_3 = \bar{x} \vee y \vee \bar{z}$

$m_4 = \bar{x} \vee y \vee z$

$m_8 = x \vee y \vee z$

$\therefore CNF = (\bar{x} \vee y \vee \bar{z}) \wedge (\bar{x} \vee y \vee z) \wedge (x \vee y \vee z)$

यसलाई meet of maxterms अथवा maxterm normal form पनि भनिन्छ।